Stack Implementation Using Array

C Program Code and Execution Output

PROGRAM SOURCE CODE

```
#include<stdio.h>

int stack[100], n, top, i;

void push(int x)
{
    if(top >= n-1)
    {
        printf("\n STACK is over flow");
    }
    else
    {
        top++;
        stack[top] = x;
        printf("\n element pushed into stack:%d", x);
    }
}

void pop()
{
    if(top <= -1)
    {
        printf("\n Stack is under flow");
    }
    else
    {
        printf("\n The popped elements is %d", stack[top]);
        stack[top] = 0;
        top--;
    }
}

void display()
{
    if(top >= 0)
    {
        printf("\n The elements in STACK:\n");
        for(i = top; i >= 0; i--)
            printf("%d \n", stack[i]);
    }
    else
    {
        printf("\n The STACK is empty");
    }
}

int main()
{
    top = -1;
    n = 5;
    push(10);
    push(20);
    push(30);
```

PROGRAM OUTPUT

```
element pushed into stack:10
element pushed into stack:20
element pushed into stack:30
element pushed into stack:40
element pushed into stack:50
 STACK is over flow
 The elements in STACK:
50
40
30
20
10

 The popped elements is 50
 The popped elements is 40
 The popped elements is 30
 The popped elements is 20
 The popped elements is 10
 Stack is under flow
```